

BUZZARDS BAY, MA, USA

WMO: 994140

Lat: 41.397N

Lon: 71.033W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.0	(c) -8.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 20.8	(k) 1.1	(l) 19.3	(m) 0.3	(n) 11.1	(o) 330	(p) 0.795

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 2.7	(c) 24.7	(d) N/A	(e) 23.8	(f) N/A	(g) 23.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.1	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.1	(b) 17.5	(c) 15.4	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(4) 20.1	(5) 17.5	(6) 15.4	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	0.2	1.2	3.4	7.6	12.2	16.8	20.8	20.8	18.3	13.7	8.5	3.8		
	DBStd	7.92	5.27	3.75	3.28	2.76	2.58	2.65	1.87	1.72	2.04	3.26	3.93	4.01		
	HDD10.0	1118	305	247	204	79	6	0	0	0	0	9	73	194		
	HDD18.3	2996	563	481	462	321	191	59	2	2	24	146	295	451		
	CDD10.0	1363	1	0	1	9	74	205	333	336	251	123	28	3		
	CDD18.3	197	0	0	0	0	1	14	77	79	24	2	0	0		
	CDH23.3	137	0	0	0	0	0	6	82	46	4	0	0	0		
	CDH26.7	7	0	0	0	0	0	0	6	1	0	0	0	0		
(14) Wind	WSAvg	7.9	9.1	9.1	9.0	8.3	7.7	6.5	6.1	5.9	6.7	8.3	8.9	9.2		
(15) Precipitation	PrecAvg	1207	97	88	118	105	89	96	87	101	106	106	98	116		
	PrecMax	1447	172	180	367	204	191	287	252	201	233	353	162	199		
	PrecMin	942	43	43	8	35	28	2	35	38	30	16	20	34		
	PrecStd	151	32	34	71	45	35	72	41	45	50	70	37	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	9.9	12.3	17.9	21.6	24.1	27.5	25.8	24.0	21.3	17.1	14.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	10.4	8.6	10.3	13.9	18.7	22.2	25.2	24.4	22.5	20.0	15.7	12.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	9.0	7.2	8.9	12.1	17.0	21.2	24.1	23.8	21.6	18.9	15.0	11.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	7.3	6.0	7.8	10.8	15.6	20.2	23.3	23.2	20.9	17.9	14.2	9.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	5.2	4.7	4.1	3.8	3.4	3.0	2.7	3.2	3.9	4.7	5.3		
		MCDBR	6.6	6.3	5.9	6.3	6.3	5.0	4.3	3.3	3.7	4.1	4.6	6.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.319	0.342	0.378	0.420	0.465	0.492	0.446	0.379	0.350	0.321	0.309		
	taud		2.502	2.483	2.452	2.382	2.302	2.215	2.149	2.271	2.434	2.515	2.556	2.552		
	Ebn at Noon		845	894	908	893	859	817	790	818	857	846	830	820		
	Edh at Noon		74	87	101	116	129	141	149	128	102	84	70	66		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.72	3.76	4.85	5.41	5.88	6.03	5.42	4.42	3.00	2.03	1.54		
	RadStd		0.15	0.27	0.28	0.52	0.52	0.57	0.32	0.30	0.33	0.27	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98	+49

Nomenclature: See separate page